Adaptive interrupt coalescing patch for NVMe SSD

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What is Adaptive Interrupt Coalescing

- Heavy small I/O stress on multiple NVMe SSDs can trigger interrupt bursts, possibly up bounding IOPS of those NVMe SSDs with high level of HW spec.

- To solve interrupt burst:
  - ➢ If IOPS continuously larger than threshold (default 500k) and running io queues num equals to or larger than threshold (default 2) in certain period (default 6 cycle with each 1 seconds), automatically enable coalescing interrupt.
  - ➢ If IOPS continuously less then threshold (default 500k) or running io queues num less than threshold (default 2) in certain period (default 6 cycle with each 1 seconds), automatically disable coalescing interrupt.

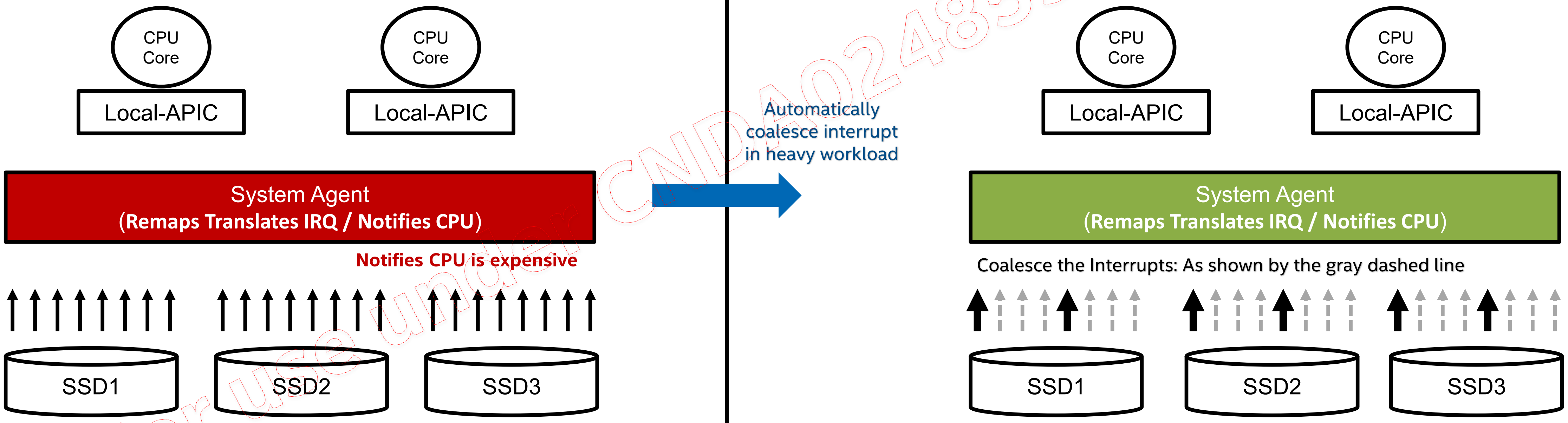
Note: 1. the check threshold 500k is a data get from experiment over multiple NVMe disks (UMIS/Solidigm/Memblaze/SAMSUNG/...). The threshold value talked above can be modified during runtime now.

How the patch works in Heavy WL

Heavy Workload Problem

(e.g. bs=4k, numjobs=8 iodepth=128)

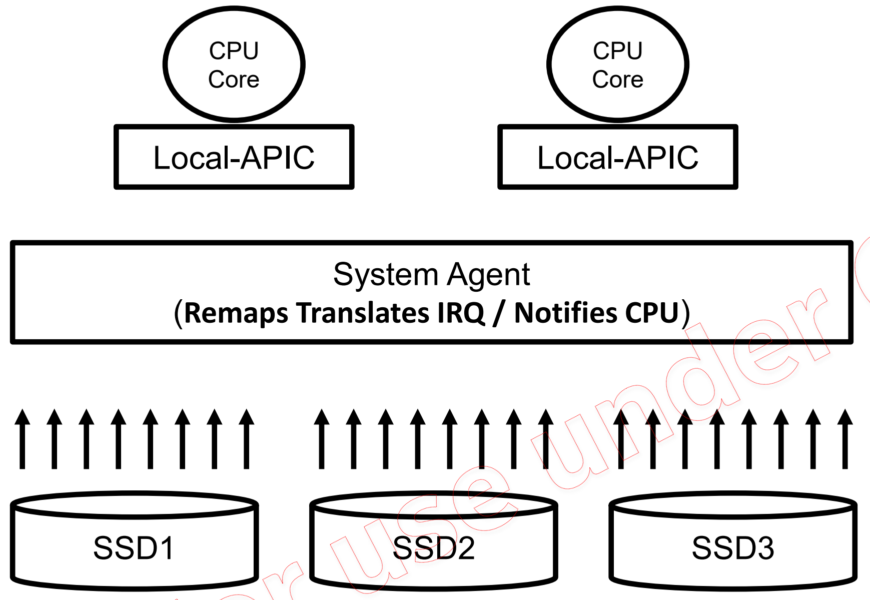
How the Patch Works – Interrupt Coalescing Enabled



✓ Interrupt handling details: https://lwn.net/Articles/951186/

How the patch works in Light WL

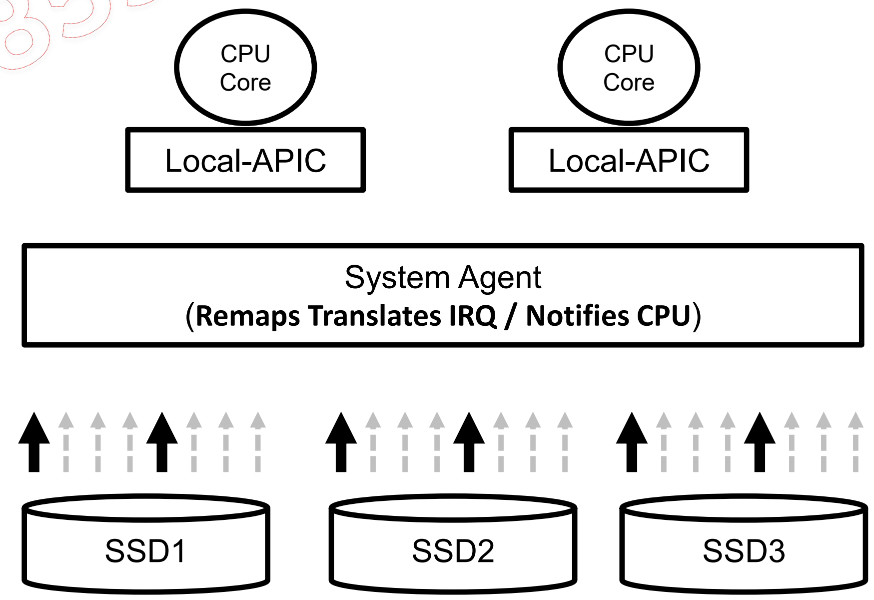
How the Patch Works – Interrupt Coalescing Disabled



Automatically disable coalesce in light Workload

Light WL

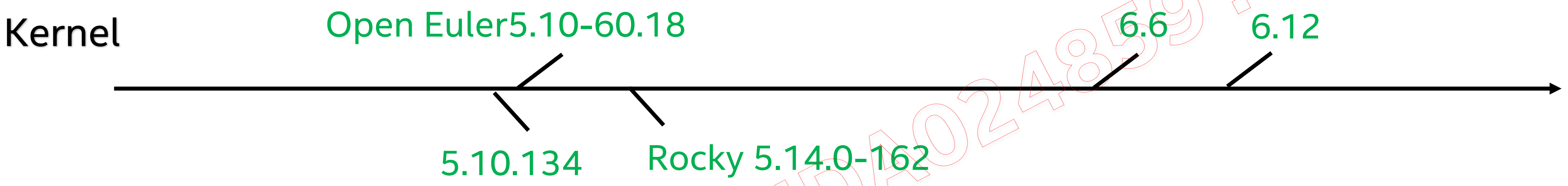
(bs=4k numjobs=1 iodepth=1 or other qd vaule)



Why low: (1) wait for requests (2) min timeout: 100us

✓ Interrupt handling details: https://lwn.net/Articles/951186/

Patches Readiness



Patch for Green version is ready

Patch Major Changes

- Description about the changes in new patch

1. use whole device statistics (IOPS & running IO queues number) to do judgement
   1) IOPS of the whole nvme device
   2) IO queues being used
2. run the checking logic in delayed work
   1) the default scheduled cycle is 1000ms and adaptive the threshold IOPS judgement according to the real time of each cycle
   2) delayed work scheduled on the last core of the specific numa which the nvme devices belong to
3. check whether the nvme device support interrupt coalescing or not during the initialization, if interrupt coalescing not supported, then not start the delayed work which used to do the check
4. not use the io size as judgement parameter avoid the fluctuations caused by multiple IO size running at the same time

Installation

❑ **Steps to install**

- git clone related Linux kernel source code
- git apply the patch
- compile Linux kernel
- make install Linux kernel
- reboot OS

❑ **How to check if installed successfully**

ls –l /sys/kernel/nvme/

If the configurations exists in this folder, it means the patch has been successfully installed.

```
[root@sh14l02SHFs0402 fiotest]# ll /sys/kernel/nvme/*
-rw-r--r--. 1 root root 4096 Nov 25 22:52 /sys/kernel/nvme/check_interval
-rw-r--r--. 1 root root 4096 Nov 25 22:52 /sys/kernel/nvme/coalesce_value
-rw-r--r--. 1 root root 4096 Nov 25 22:52 /sys/kernel/nvme/disable
-rw-r--r--. 1 root root 4096 Nov 25 19:43 /sys/kernel/nvme/high_io_delta
-rw-r--r--. 1 root root 4096 Nov 25 22:52 /sys/kernel/nvme/hit_count_before_change
-rw-r--r--. 1 root root 4096 Nov 25 22:52 /sys/kernel/nvme/min_running_queues
-rw-r--r--. 1 root root 4096 Nov 25 19:43 /sys/kernel/nvme/scan_interval
-rw-r--r--. 1 root root 4096 Nov 25 22:52 /sys/kernel/nvme/switch_cnt
```

Configuration

*Don't change the value before you really know what you are doing.

❑ configurations are located in /sys/kernel/nvme/

Parameters	Default	Descriptions
coalesce_value	266	The coalesce value set to NVME SSD, similar to below command: *nvme set-feature /dev/nvme$i -f 0x8 --value=266 -n 0* 266 == 0x10a, that means timeout is set to 100us, coalesced request count is set to 0xa. If want to set 0x109, just need to “echo 265 > /sys/kernel/nvme/coalesce_value”
disable	0	0: enable, to enable adaptive interrupt coalescing, “echo 0 > /sys/kernel/nvme/disable” 1: disable, to disable adaptive interrupt coalescing, “echo 1 > /sys/kernel/nvme/disable”
hit_count_before_change	5	How many consecutive detection cycles of nvme device busy should be satisfied before changing coalescing status between coalescing and non-coalescing. See more description in high_io_delta and min_running_queues
scan_interval	1000	the scheduled period of delayed work which used to do the checking and judgement, with unit in ms. When changing the scan_interval, the high_io_delta will automatically adjust the high_io_delta threshold according to this setting value.
high_io_delta	500000	IOPS per nvme device which used to judge whether to set the interrupt coalescing. When IOPS reach or larger than this value and running IO queues larger than the min_running_queues value then mark the nvme device as busy for 1 time else then mark as non-busy for 1 time (will clear the other when 1 is matched). And when the busy time or non-busy time reach the above hit_count_before_change threshold will trigger the operation of setting or clearing the interrupt coalescing feature.
min_running_queues	2	The min num of IO queues been used during each check cycle (see scan_interval above). The running IO queues need to equal or larger than this value when want to set the interrupt coalescing.
check_interval	512	the IO interval to mark whether a specific IO queue of a specific nvme device is been used or not.
switch_cnt	0	Indicate how many time the status has switched between coalescing and non-coalescing. echo 1 > disable will reset this value to 0.

Patch with 16x UMIS UH812a PCIe Gen5 Reference Performance Data

容量	规格/Intel平台16盘实测	随机读（4k)(QD=1,job=1)		随机读（4k)(QD=32,job=1)		随机写（4k)(QD=1,job=1)		随机写（4k)(QD=32,job=1)		随机混合读写（4k)(QD=16,job=1)	
		99%	99.99%	99%	99.99%	99%	99.99%	99%	99.99%	r_99.9%	w_99.9%
7680G	单盘规格	85	85	150	210	10	20	160	210	400	400
	kernel										
	Intel平台16盘实测-1	61	71.5	95.56	141.75	5.31	13.9	73	81.69	336.56	15.94
	Intel平台16盘实测-2	61	71.44	95.62	141	5.29	13.88	59.95	70.51	321.75	15.95
	Intel平台16盘实测-3	61	71.5	95.5	140.75	5.28	13.82	60.03	70.66	334.25	16.02
	3轮平均值	61	71.48	95.56	141.1666667	5.293333333	13.8666667	64.3266667	74.28666667	330.8533333	15.97
	3轮平均值-规格达成	139.34%	118.91%	156.97%	148.76%	188.92%	144.23%	248.73%	282.69%	120.90%	2504.70%

测试FW版本：1092 (UMIS 最近更新的版本，解决IO运行过程中配置聚合概率不生效问题), FIO: thread=0
2路 GNR-SP 64C, HT on, Kernel: 6.6 + 自适应中断聚合Patch, FIO 配合numactrl --localalloc + 绑核 (FIO cpus_allowed参数)

容量	规格/Intel平台16盘实测	IO场景								
		顺序读（128k)-MB/s	顺序写（128k)-MB/s	随机读IOPS（4k)-k	随机写IOPS（4k)-k	随机混合读写IOPS（4k70读30写)-k	随机读QD1时延（4k)-us	随机写QD1时延（4k)-us	顺序读QD1时延（4k)-us	顺序写QD1时延（4k)-us
7680G	单盘规格	14800	10000	3100	410	696	55	9	8	9
	16盘规格	236800	160000	49600	6560	11136	55	9	8	9
	Kernel									
	Intel平台16盘实测-1	238400	169600	55,108	7,691	19027	53.12	5.64	6.83	5.63
	Intel平台16盘实测-2	238400	169600	55,118	9,037	21435	53.11	5.64	6.83	5.63
	Intel平台16盘实测-3	238400	169600	55,112	9, 063	22130	53.10	5.64	6.83	5.63
	3轮平均值	238400	169600	55,116	8,597	20864	53.11	5.64	6.83	5.63
	3轮平均值-规格达成	100.68%	106.00%	111.2%	131.05%	187.36%	103.56%	159.57%	117.13%	159.86%

测试FW版本：1092 （UMIS最近更新的版本） 每个盘片绑八个核，不跨node绑核 (numactl --localalloc)，thread=1，测试前先跑2遍4K随机写数据预埋
2路 GNR-SP 64C, HT on, Kernel: 6.6 + 自适应中断聚合Patch, FIO 配合numactrl --localalloc + 绑核 (FIO cpus_allowed参数)

NextSetp

1. **Do more test and target for upstream**